PROJECT TITLE

Bike Sharing Network Performance Analysis Using Power BI

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1. Project Overview and Objective

Project Overview

This project focuses on analysing a public bike-sharing network using Excel and Power BI. The dataset contains information about bike stations, bike availability, docking capacity, operational status, and contract locations. The raw dataset was cleaned and transformed in Power query before being imported into Power BI for data modelling and interactive dashboard creation.

Objective

The primary objective of this project is to monitor bike station performance, analyse bike availability across different contracts, evaluate station operational status, and generate meaningful business insights to support efficient fleet management and improve customer experience.

2. Data Source

Source: Public Bike Sharing Dataset

Dataset Features

- Station Details
- Contract Name (City)
- Station Status
- Available Bikes
- Available Bike Stands
- Bike Stand Capacity
- Last Updated Timestamp
- Geographic Coordinates

3. Problem Statement

The bike-sharing company operates multiple stations across different cities. Efficient monitoring of station availability and bike distribution is essential for providing uninterrupted service to customers.

The project aims to answer the following business questions:

- Which contracts have the highest bike availability?
- How many stations are currently operational?
- Which stations require operational attention?
- How are bikes distributed across different contracts?
- What is the overall performance of the bike-sharing network?

4. Attribute Details

ATTRIBUTE	DATA TYPE	DESCRIPTION
NUMBER	WHOLE NUMBER	UNIQUE STATION ID
NAME	TEXT	BIKE STATION NAME
ADDRESS	TEXT	STATION ADDRESS
POSITION	GEOGRAPHIC	LATITUDE&LONGITUDE
BANKING	TEXT	BANKING FACILITY AVAILABLE
BONUS	TEXT	BONUS STATION INDICATOR
STATUS	TEXT	STATION STATUS
CONTRACT NAME	TEXT	CITY/CONTRACT
BIKE STAND	WHOLE NUMBER	TOTAL BIKE STAND
AVAILABLE BIKE STANDS	WHOLE NUMBER	AVAILABLE BIKE STAND
AVAILABLE BIKE	WHOLE NUMBER	AVAILABLE BIKE AT STATION
LAST UPDATE	DATE TIME	LAST UPDATED DATE&TIME

5. Tools & Technologies

POWER QUERY

- Data Cleaning
- Remove Duplicates
- Data Formatting
- Data Validation

Power BI

- Power Query
- Data Modelling
- DAX Measures
- Interactive Dashboard
- KPI Cards
- Slicers
- Charts & Visualizations

6. Data Pre-Processing (Excel / Power Query)

The following preprocessing steps were performed before analysis:

- Imported raw Excel dataset into Power BI
- Verified data types
- Removed duplicate records
- Standardized column names
- Converted Date column into Date format
- Created Fact and Dimension tables
- Built relationships between tables
- Loaded cleaned data into Power BI

7. Data Modelling and DAX (Power BI)

Data Model

A Star Schema model was created using Fact and Dimension tables.

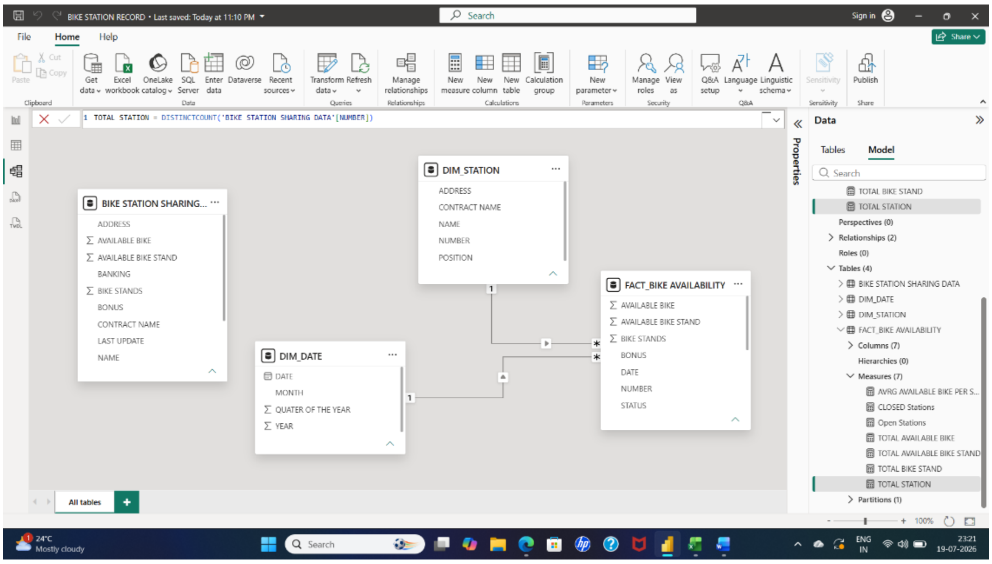
Dimension Tables

- DIM_DATE
- DIM_STATION

Fact Table

- FACT_BIKE_AVAILABILITY

Relationships were established using Station Number and Date fields.



DAX Measures

- Total Stations
- Total Records
- Total Bike Stands
- Total Available Bike Stands
- Total Bikes
- Total Available Bikes
- Open Stations
- Closed Stations

8. Analysis and Visualizations (Power BI)

KPI Cards

- Total Stations
- Total Bike Stands
- Total Available Bikes
- Total Available Bike Stands
- Open Stations
- Closed Stations

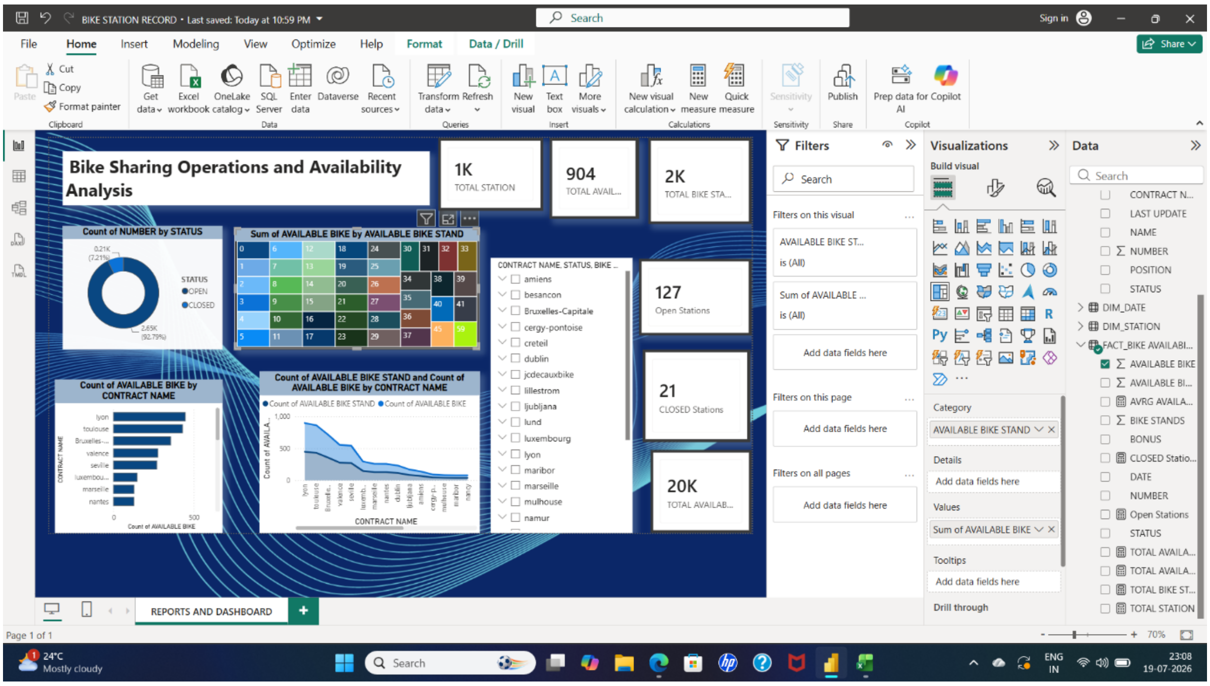
Charts

- Donut Chart – Station Status Distribution
- Bar Chart – Contract-wise Bike Availability
- Line Chart – Bike Availability by Contract

Interactive Features

- Contract Name Slicer
- Cross Filtering
- Dynamic Dashboard Interaction

Dashboard Features



9. Insights & Conclusions

Key Findings

- Most bike stations are operational.
- A few stations are currently closed.
- Bike availability varies across different contracts.

- Some contracts have significantly higher bike availability.

- Bike distribution is not uniform across all contracts.
- Certain contracts require better bike allocation.
- Closed stations reduce overall service availability.

- Contracts with consistently lower bike availability may experience higher customer demand.
- Future bike allocation can be optimized using historical availability trends.

- Increase bike allocation in contracts with lower availability.
- Improve monitoring of closed stations.
- Perform periodic redistribution of bikes across stations.
- Optimize station operations based on contract performance.

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10. Conclusion

This project successfully demonstrates the end-to-end data analysis process using Power query and Power BI. The dashboard provides a comprehensive view of bike-sharing operations by monitoring station status, bike availability, and contract-wise performance. The insights generated can support better operational planning, efficient fleet distribution, and improved customer service.